Baseball Statcast
Full Model Report (2019–2024)

This report covers all six target variables: **TB, BB, AB, H** (linear models) and **SB, CS** (count/nonlinear models). Parts 3 and 4 present expected values and a mathematical comparison of **MB vs xMB**. Models are only deployed if $R^2 \geq$ **0.8**; otherwise the expected value equals the actual observed value.

Overall Best Model Summary

Target	Best Model	R^2	CV R^2	RMSE	Deployed?
TB	Lasso Regression	0.9542	0.9484 ± 0.0128	17.1030	Yes
BB	Random Forest	0.8992	0.8934 ± 0.0035	6.6596	Yes
AB	Random Forest	0.9758	0.9766 ± 0.0018	24.6978	Yes
H	Ridge Regression	0.9379	0.9366 ± 0.0056	11.2421	Yes
SB	Poisson GLM (5 features)	0.4099	0.405	7.95	No (actual used)
CS	Neg. Binomial GLM (5 features)	0.2863	0.2804	2.2469	No (actual used)

Part 1 — Regression Targets: TB, BB, AB, H

Linear, Ridge, Lasso, and Random Forest models were evaluated. Features selected by |correlation| > 0.3 with each target.

Target: TB

Selected features: Barrel%, maxEV, HardHit, HardHit%, CSW%, EV, Pitches

Model	R^2	CV R^2	RMSE	Alpha
Linear Regression	0.9541	0.9484 ± 0.0126	17.1072	N/A
Ridge Regression	0.9541	0.9484 ± 0.0127	17.1072	0.1000
Lasso Regression	**0.9542**	**0.9484 ± 0.0128**	**17.1030**	**0.0765**
Random Forest	0.9463	0.9421 ± 0.0142	18.5082	N/A

Lasso Coefficients

Feature	Coefficient
HardHit	55.3379
Pitches	23.5446
HardHit%	-16.0030
Barrel%	15.5665
CSW%	-2.7247
EV	1.3924
maxEV	0.0000

Random Forest Feature Importances

Feature	Importance
Pitches	0.5657
HardHit	0.3882
Barrel%	0.0159
HardHit%	0.0093
maxEV	0.0077
CSW%	0.0073
EV	0.0060

Target: BB

Selected features: O-Swing%, Swing%, F-Strike%, Barrel%, maxEV, HardHit, HardHit%, EV, Pitches

Model	R^2	CV R^2	RMSE	Alpha
Linear Regression	0.8758	0.8581 ± 0.0064	7.3930	N/A
Ridge Regression	0.8757	0.8581 ± 0.0064	7.3954	1.0000
Lasso Regression	0.8757	0.8577 ± 0.0063	7.3977	0.0178
Random Forest	**0.8992**	**0.8934 ± 0.0035**	**6.6596**	**N/A**

Lasso Coefficients

Feature	Coefficient
Pitches	18.0111
O-Swing%	-5.2224
Barrel%	2.2085
HardHit	-1.7825
F-Strike%	-1.2778
Swing%	-0.7874
maxEV	0.5598
EV	0.4970
HardHit%	-0.1214

Random Forest Feature Importances

Feature	Importance
Pitches	0.7523
O-Swing%	0.1252
F-Strike%	0.0397
Barrel%	0.0213
Swing%	0.0192
maxEV	0.0131
EV	0.0109
HardHit%	0.0095
HardHit	0.0087

Target: AB

Selected features: maxEV, HardHit, CSW%, Pitches

Model	R^2	CV R^2	RMSE	Alpha
Linear Regression	0.9735	0.9733 ± 0.0018	25.8457	N/A
Ridge Regression	0.9734	0.9733 ± 0.0018	25.8533	1.0000
Lasso Regression	0.9735	0.9733 ± 0.0018	25.8190	0.1568
Random Forest	**0.9758**	**0.9766 ± 0.0018**	**24.6978**	**N/A**

Lasso Coefficients

Feature	Coefficient
Pitches	122.1252
HardHit	38.1429
maxEV	-6.3322
CSW%	-5.5592

Random Forest Feature Importances

Feature	Importance
Pitches	0.9739
HardHit	0.0129
CSW%	0.0072
maxEV	0.0060

Target: H

Selected features: maxEV, HardHit, CSW%, Pitches

Model	R^2	CV R^2	RMSE	Alpha
Linear Regression	0.9379	0.9366 ± 0.0055	11.2426	N/A
Ridge Regression	**0.9379**	**0.9366 ± 0.0056**	**11.2421**	**1.0000**
Lasso Regression	0.9379	0.9366 ± 0.0056	11.2425	0.0430
Random Forest	0.9362	0.9323 ± 0.0067	11.3973	N/A

Lasso Coefficients

Feature	Coefficient
Pitches	22.3248
HardHit	22.2230
CSW%	-3.7652
maxEV	-2.7972

Random Forest Feature Importances

Feature	Importance
Pitches	0.6809
HardHit	0.2790
CSW%	0.0231
maxEV	0.0169

Part 2 — Count Targets: SB, CS

SB and CS are non-negative integer counts with nonlinear relationships to speed metrics. Six model families were tested: Linear, Sqrt transform, Log transform, Polynomial deg-2, Power Law / Negative Exponential, and a count GLM (Poisson for SB; Neg. Binomial for CS).

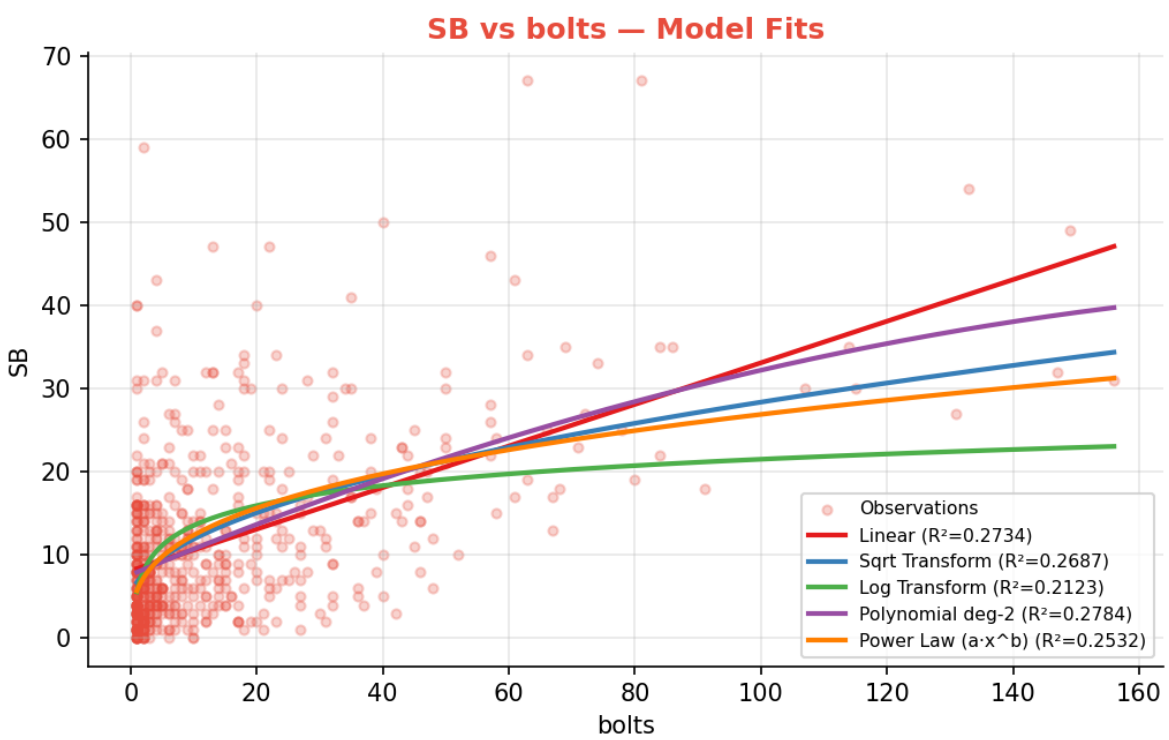
Target: SB

Primary predictor: **bolts**

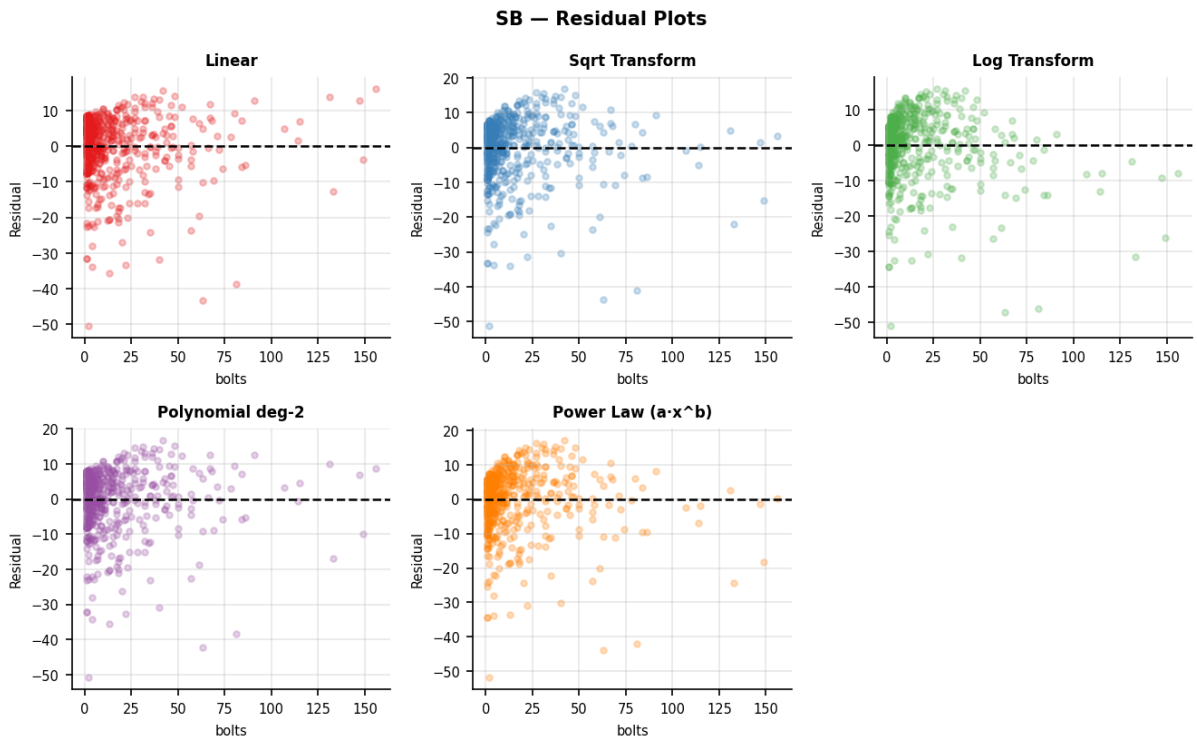
Model	R^2	Adj R^2	RMSE	Notes
Poisson GLM (5 features)	**0.4099**	**0.405**	**7.95**	**Count GLM — log-link, multi-feature**
Polynomial deg-2	0.2784	0.276	8.7544	Quadratic
Linear	0.2734	0.2722	8.7848	Baseline
Sqrt Transform	0.2687	0.2676	8.8127	SB ~ sqrt(bolts)
Power Law (a·x^b)	0.2532	0.2508	8.9057	a=5.697, b=0.337
Log Transform	0.2123	0.211	9.1467	SB ~ log(bolts)

Green row = best R^2.

Fitted Curves



Residual Plots



Poisson GLM (5 features) — Coefficients

Feature	Coefficient
const	2.3190
Pitches	0.5683
sprint_speed	0.1674
HardHit	-0.1375
bolts	0.0987
hp_to_1b	-0.0853

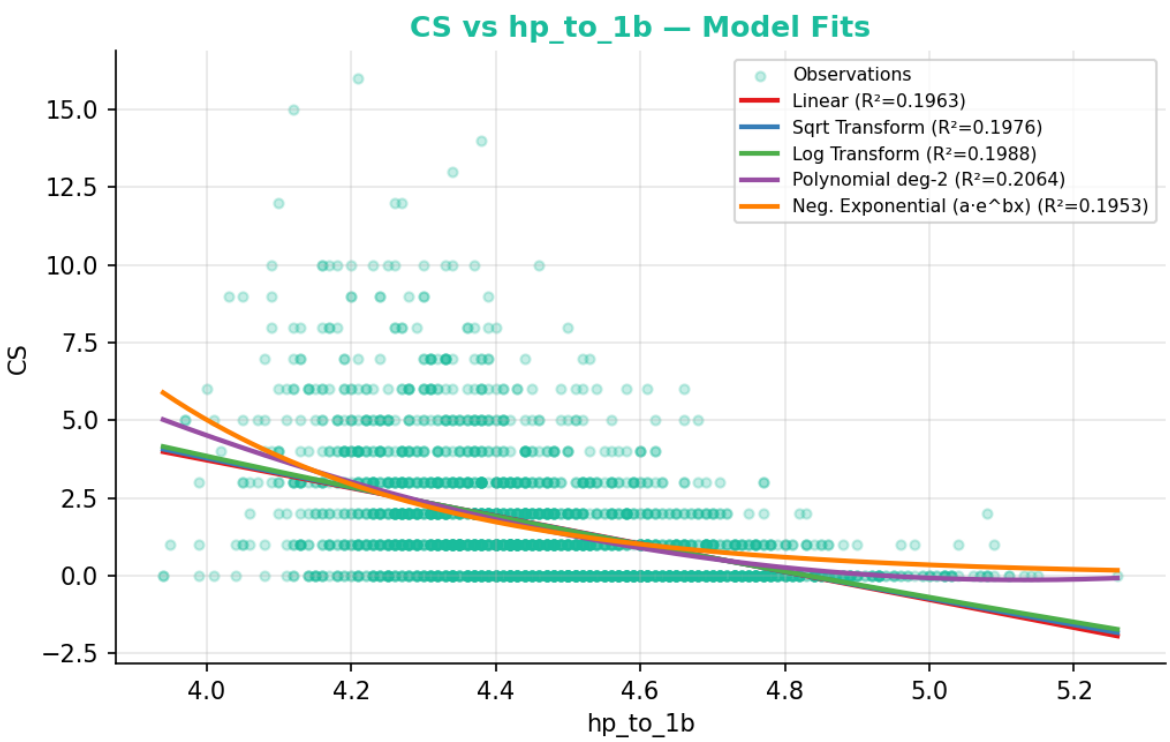
Target: CS

Primary predictor: **hp_to_1b**

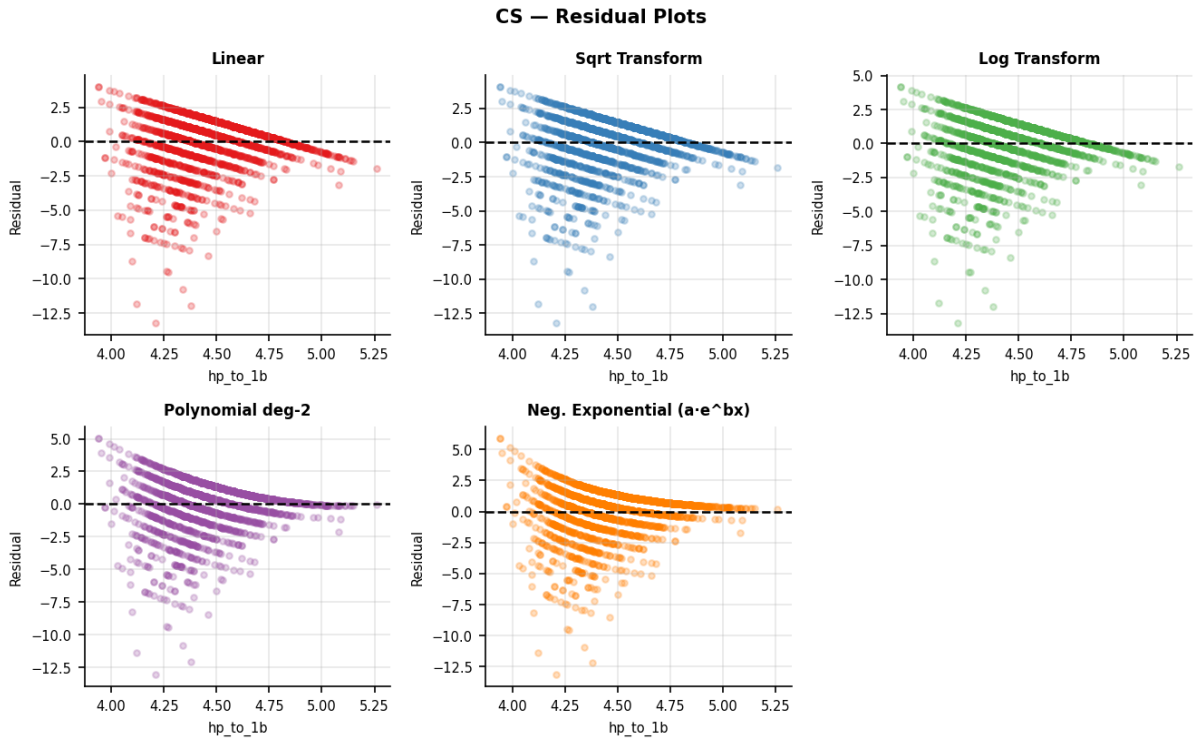
Model	R^2	Adj R^2	RMSE	Notes
Neg. Binomial GLM (5 features)	**0.2863**	**0.2804**	**2.2469**	**Count GLM — log-link, multi-feature**
Polynomial deg-2	0.2064	0.2057	1.8476	Quadratic
Log Transform	0.1988	0.1984	1.8564	CS ~ log(hp_to_1b)
Sqrt Transform	0.1976	0.1973	1.8578	CS ~ sqrt(hp_to_1b)
Linear	0.1963	0.196	1.8593	Baseline
Neg. Exponential (a·e^bx)	0.1953	0.1946	1.8604	a=211044.808, b=-2.662

Green row = best R^2.

Fitted Curves



Residual Plots



Neg. Binomial GLM (5 features) — Coefficients

Feature	Coefficient
const	0.9921
Pitches	0.5645
HardHit	-0.1519
sprint_speed	0.1372
hp_to_1b	-0.0613
bolts	0.0575

Part 3 — Expected Values (Model Deployment)

Best models with $R^2 \geq 0.8$ were deployed to calculate expected values. Where no model met the threshold, the expected value equals the actual observed value. Full results are saved to **expected_all_stats.csv**.

Name	Sea son	TB	xTB	BB	xBB	AB	xAB	H	xH	SB	xSB	CS	xCS	MB	xM
Tim Locastro	2019.0	72.0	68.75	14.0	14.83	212.0	210.56	53.0	42.77	17.0	17.0	0.0	0.0	25.0	21
Trea Turner	2019.0	259.0	226.88	43.0	44.73	521.0	510.62	155.0	133.47	35.0	35.0	5.0	5.0	96.0	78
Byron Buxton	2019.0	139.0	108.79	19.0	19.11	271.0	259.19	71.0	61.17	14.0	14.0	3.0	3.0	44.0	33
Myles Straw	2019.0	37.0	45.07	19.0	18.9	108.0	119.56	29.0	30.55	8.0	8.0	1.0	1.0	16.0	18
Roman Quinn	2019.0	40.0	27.89	12.0	11.8	108.0	105.25	23.0	20.82	8.0	8.0	0.0	0.0	13.0	9.
Garrett Hampson	2019.0	115.0	100.19	24.0	27.77	299.0	298.0	74.0	63.32	15.0	15.0	3.0	3.0	37.0	30
Tyler O'Neill	2019.0	58.0	41.34	10.0	10.19	141.0	138.2	37.0	21.08	1.0	1.0	0.0	0.0	18.0	9.
Adam Engel	2019.0	87.0	75.75	14.0	14.92	227.0	223.41	55.0	43.34	3.0	3.0	3.0	3.0	24.0	18
dalberto Mondesi	2019.0	176.0	154.26	19.0	19.59	415.0	393.0	109.0	86.17	43.0	43.0	7.0	7.0	57.0	43
Jon Berti	2019.0	104.0	84.89	24.0	25.38	256.0	252.09	70.0	57.31	17.0	17.0	3.0	3.0	37.0	27
Harrison Bader	2019.0	127.0	144.42	46.0	43.91	347.0	357.99	71.0	81.28	11.0	11.0	3.0	3.0	39.0	46
Billy Hamilton	2019.0	87.0	87.25	32.0	32.52	316.0	312.9	69.0	60.22	22.0	22.0	6.0	6.0	27.0	25
Richie Martin	2019.0	91.0	90.77	14.0	16.02	283.0	278.51	59.0	60.97	10.0	10.0	1.0	1.0	24.0	25
Lewis Brinson	2019.0	50.0	71.55	13.0	13.46	226.0	220.72	39.0	45.93	1.0	1.0	1.0	1.0	11.0	18
Kevin Kiermaier	2019.0	178.0	176.32	26.0	26.13	447.0	433.57	102.0	107.77	19.0	19.0	5.0	5.0	49.0	53
elino DeShields	2019.0	124.0	119.77	38.0	40.48	357.0	370.56	89.0	83.04	24.0	24.0	6.0	6.0	42.0	38
Tommy Edman	2019.0	163.0	129.75	16.0	20.41	326.0	321.72	99.0	80.31	15.0	15.0	1.0	1.0	58.0	41
Mallex Smith	2019.0	171.0	155.56	42.0	44.95	510.0	513.83	116.0	103.4	46.0	46.0	9.0	9.0	54.0	46
Nick Senzel	2019.0	160.0	155.34	30.0	32.84	375.0	383.1	96.0	95.42	14.0	14.0	5.0	5.0	50.0	48
Ryan Cordell	2019.0	77.0	81.93	19.0	18.36	217.0	215.42	48.0	47.94	3.0	3.0	1.0	1.0	22.0	23
ernando Tatis Jr.	2019.0	197.0	147.37	30.0	31.21	334.0	326.39	106.0	74.01	16.0	16.0	6.0	6.0	73.0	43
Victor Robles	2019.0	229.0	184.43	35.0	39.22	546.0	543.27	139.0	112.25	28.0	28.0	9.0	9.0	70.0	49
Trevor Story	2019.0	326.0	262.36	58.0	59.55	588.0	587.19	173.0	159.3	23.0	23.0	8.0	8.0	117.0	90
Mike Trout	2019.0	303.0	264.05	110.0	107.27	470.0	500.07	137.0	135.67	11.0	11.0	2.0	2.0	131.0	101
Keon Broxton	2019.0	56.0	57.5	20.0	22.6	204.0	208.23	34.0	30.7	10.0	10.0	6.0	6.0	13.0	13
Amed Rosario	2019.0	266.0	250.79	31.0	31.93	616.0	608.04	177.0	151.44	19.0	19.0	10.0	10.0	84.0	70
Jake Marisnick	2019.0	120.0	97.29	17.0	18.61	292.0	287.15	68.0	61.91	10.0	10.0	3.0	3.0	33.0	26

Name	Sea son	TB	xTB	BB	xBB	AB	xAB	H	xH	SB	xSB	CS	xCS	MB	xM
Charlie Tilson	2019.0	41.0	41.51	10.0	10.16	144.0	139.79	33.0	31.6	4.0	4.0	0.0	0.0	12.0	12
ory Spangenberg	2019.0	34.0	36.66	6.0	5.65	95.0	94.54	22.0	15.99	3.0	3.0	0.0	0.0	10.0	8.
Scott Kingery	2019.0	217.0	178.1	34.0	36.22	458.0	442.16	118.0	105.55	15.0	15.0	4.0	4.0	68.0	54

Showing first 30 rows. Full results saved to expected_all_stats.csv.

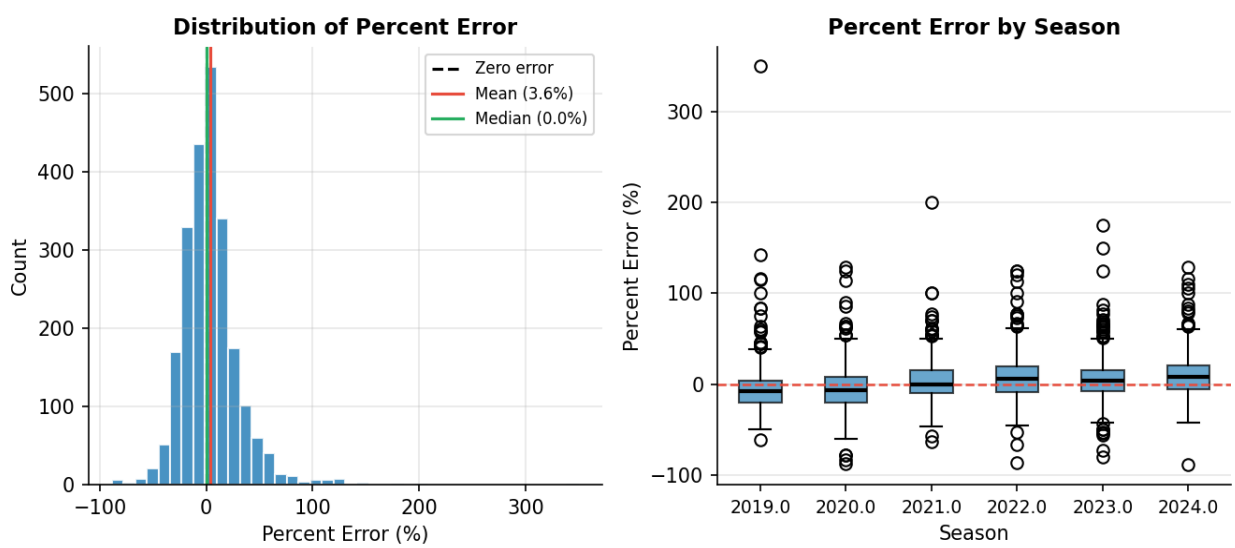
Part 4 — MB vs xMB (Percent Error & Bayesian Comparison)

This section evaluates how well **xMB** — derived from ML predictions of individual counting stats — matches the **actual MB** computed from observed statistics. Two complementary tests are applied: a **Percent Error analysis** measuring prediction accuracy at the player level, and a **Bayesian Model Comparison** assessing whether xMB carries statistically meaningful predictive information about MB beyond a simple mean baseline.

Percent Error Summary

Metric	Value
Players analysed	2325
Mean Percent Error	3.64% (+ = over-predict)
Median Percent Error	0.0%
Mean Abs. Percent Error	18.66%
Median Abs. Percent Error	13.64%
Std Dev of Percent Error	27.16%
Min Percent Error	-88.89%
Max Percent Error	350.0%
Players within ±5%	20.7%
Players within ±10%	38.9%
Players within ±20%	67.4%

Error Distribution & Seasonal Breakdown



Left: Distribution of percent error across all player-seasons. Right: Box plots of percent error by season.

Top 5 Most Over-Predicted Players (xMB >> MB)

Name	Season	MB	xMB	% Error
Zack Cozart	2019	2.0	9.0	350.0%
Jonathan Davis	2021	4.0	12.0	200.0%
Cam Gallagher	2023	4.0	11.0	175.0%
Austin Nola	2023	6.0	15.0	150.0%
Josh Harrison	2019	7.0	17.0	142.9%

Top 5 Most Under-Predicted Players (xMB << MB)

Name	Season	MB	xMB	% Error
Oliver Dunn	2024	9.0	1.0	-88.9%
Joey Bart	2020	8.0	1.0	-87.5%
David Bote	2022	15.0	2.0	-86.7%
Taylor Ward	2020	12.0	2.0	-83.3%
Everson Pereira	2023	5.0	1.0	-80.0%

Bayesian Model Comparison

Two models of MB are compared using the **Bayesian Information Criterion (BIC)**. The **Bayes Factor (BF_{10})** is derived as $\exp((BIC_0 - BIC_1) / 2)$ and interpreted on Jeffreys' scale.

Model	Description	Parameters
Model 0 (Null)	MB ~ mean(MB) — xMB has no predictive value	1 (mean)
Model 1 (xMB)	MB ~ a + b·xMB — xMB linearly predicts MB	2 (intercept, slope)

Statistic	Value
BIC — Model 0 (Null)	15670.94
BIC — Model 1 (xMB predictor)	10063.49
Delta BIC (BIC₀ – BIC₁)	5607.45
Bayes Factor (BF₁₀)	inf
log₁₀ Bayes Factor	inf
Posterior P(Model 1 | data)	nan%
OLS Intercept (a)	0.46
OLS Slope (b)	0.995
Model 1 R<super>2</super>	0.9106

Verdict (Jeffreys' Scale): Decisive evidence for Model 1

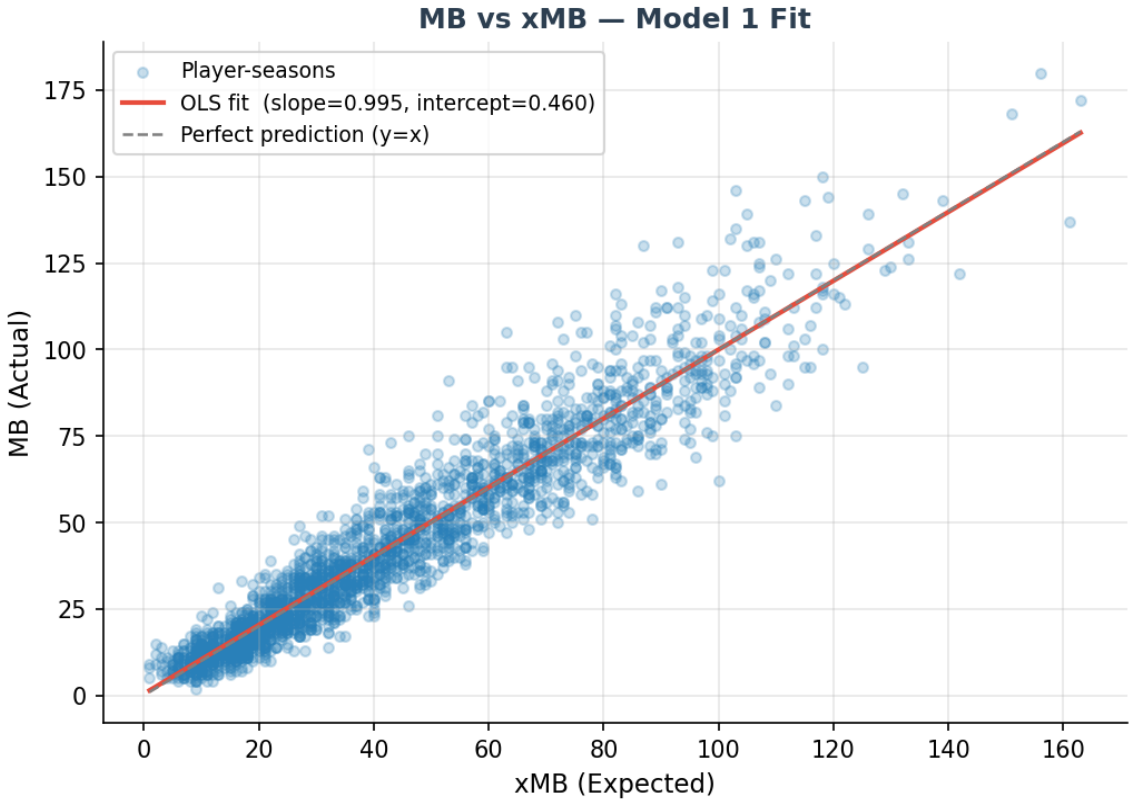
Jeffreys' Scale Reference

Bayes Factor (BF_{10})	Interpretation
< 1	Against Model 1 (xMB uninformative)
1 – 3	Anecdotal evidence for Model 1
3 – 10	Moderate evidence for Model 1
10 – 30	Strong evidence for Model 1
30 – 100	Very Strong evidence for Model 1
> 100	**Decisive evidence for Model 1**

Green row = where your result falls on the scale.

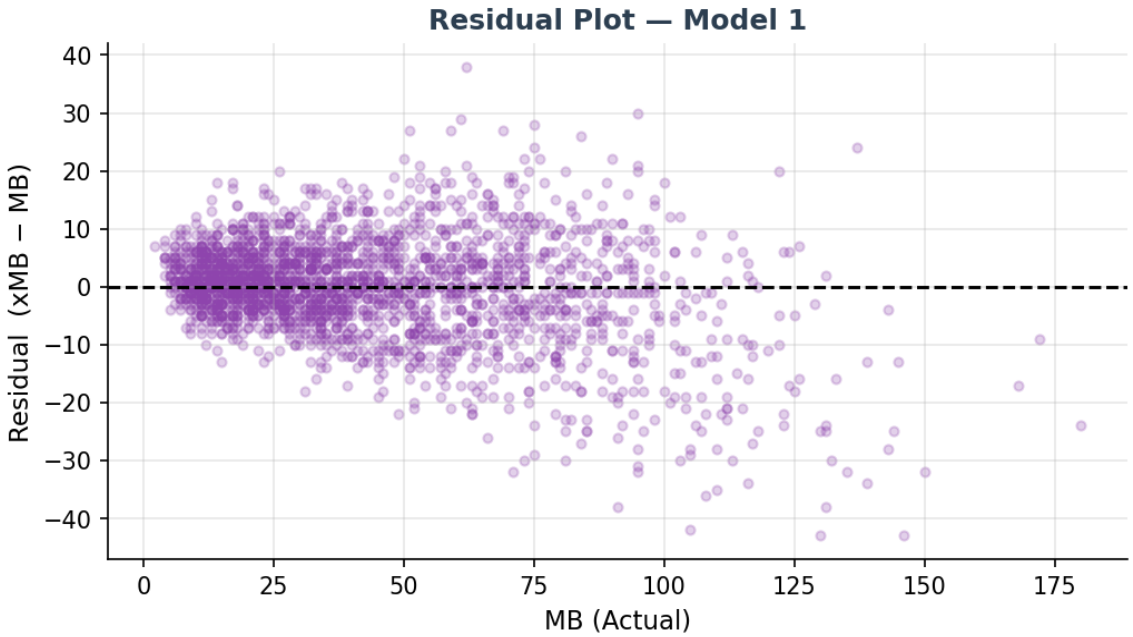
Model Visualisations

MB vs xMB Scatter (Model 1 OLS Fit)



Each point is one player-season. Red line = OLS fit; dashed grey = perfect prediction (y=x). A slope near 1 and intercept near 0 indicates good calibration.

Residual Plot (xMB – MB vs MB)



Residuals should be randomly scattered around zero. A funnel shape indicates heteroskedasticity; a curve suggests a non-linear relationship.